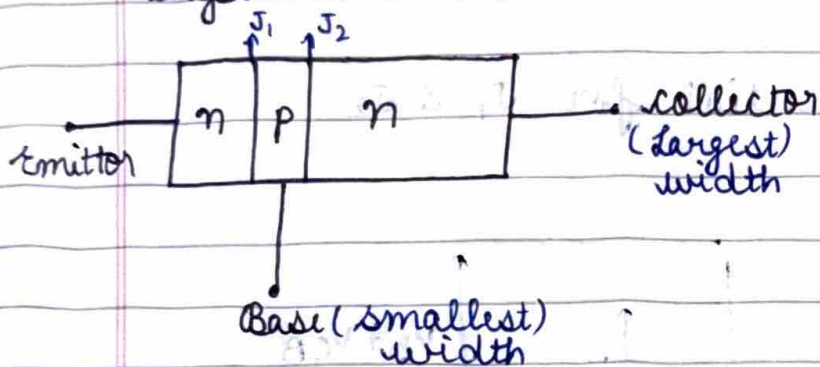


unit - 2

BIPOLAR JUNCTION TRANSISTOR

BJT is 3 terminal device used to amplify weak signals and switching operations.

→ Physical Structure

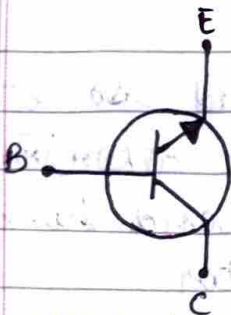


$J_1 \rightarrow$ emitter base junction

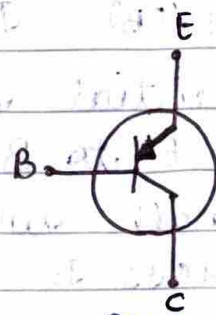
$J_2 \rightarrow$ base collector junction

width : $C > E > B$

doping : $E > C > B$



NPN transistor

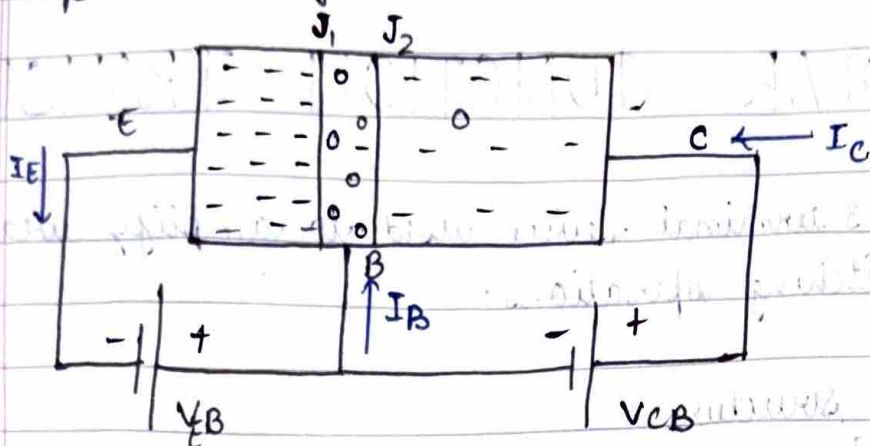


PNP transistor

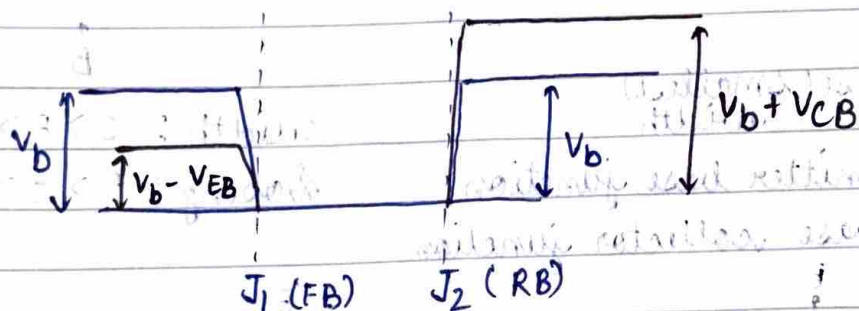
→ Regions of operation

J_1	J_2	Region of op	Feature
FB	RB	Active	Works as amp
FB	FB	saturated	on
RB	RB	cut off	off
RB	FB	Inverted	Rarely used

→ operation of Transistor



V_b → Barrier potential for J_1 & J_2



Barrier potential of J_1 is reduced so e^- move from E to B & very few recombine due to small width of base and hence most e^- cross J_2 & reach collector

(2-5%) recombine with holes at B and 95% reach collector

$$I_C = \alpha I_E + I_{CBO}$$

Small current b/t C & B

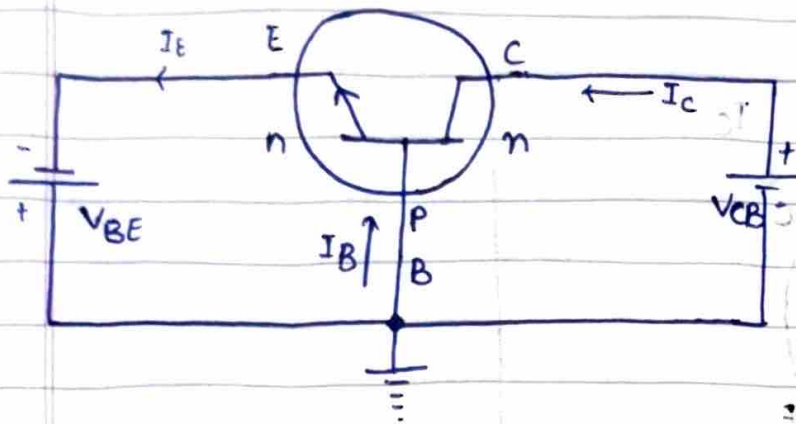
$$I_E = I_B + I_C$$

Let $(1+\alpha)N$ e^- moves from E to B, N e^- recombine at B and αN reaches C

This α is current gain factor

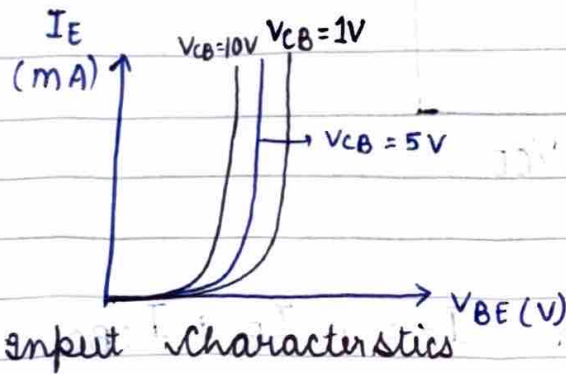
$$\alpha \in (0.95, 0.98) \approx 1$$

→ Common Base configuration (B is common to input & output ckt and is grounded)

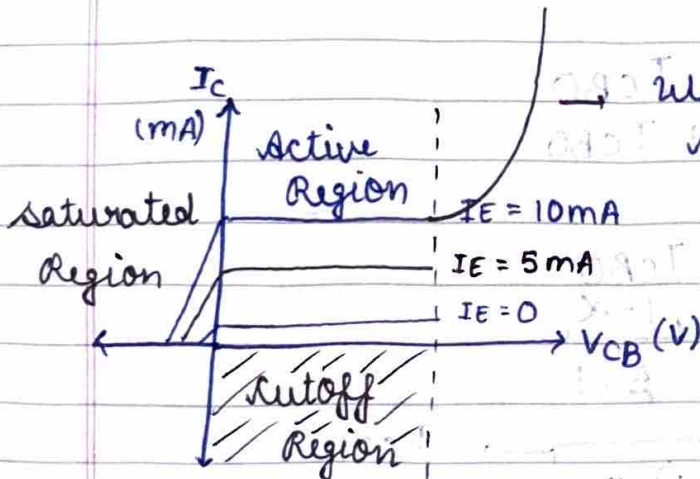


Active mode

$J_1 \rightarrow EB$ is Forward
& $J_2 \rightarrow BC$ is Reverse Biased



Input Characteristics



Output Characteristics

Working Principle (NPN)

When J_1 is F.B., electrons move from E to B. B is thin & lightly doped so only a few recombine & remaining move to collector.

$$\alpha = \frac{I_C}{I_E}$$

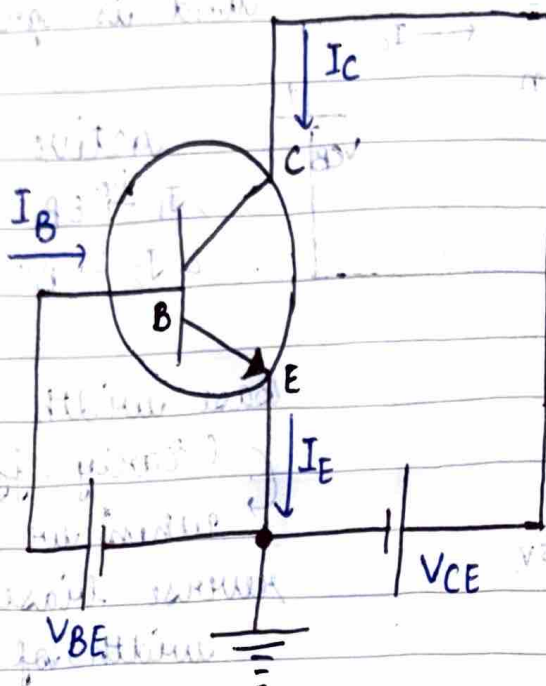
Current gain

Base width modulation (Early effect)

when we increase reverse biased V_{CB} width of base modulates & $\uparrow \uparrow$ conc gradient $\uparrow \uparrow$

When we increase V_{CB} very much, diode breaks and I_C but this situation never arises because the transistor can't withstand large amount of current.

→ Common emitter configuration of transistor



$$I_E = I_B + I_C$$

$$I_C = \alpha I_E + I_{CBO}$$

$$I_C = \alpha (I_B + I_C) + I_{CBO}$$

$$(1 - \alpha) I_C = \alpha I_B + I_{CBO}$$

$$I_C = \frac{\alpha}{1 - \alpha} I_B + \frac{I_{CBO}}{1 - \alpha}$$

$$I_C = \beta I_B + (\beta + 1) I_{CBO}$$

$$I_C = \beta I_B + I_{CEO}$$

$$I_{CEO} \ll \beta I_B$$

$$\Rightarrow I_C = \beta I_B$$

$$\Rightarrow \beta = \frac{I_C}{I_B}$$

→ Current amplification factor $\beta \in (50, 100)$

For common base

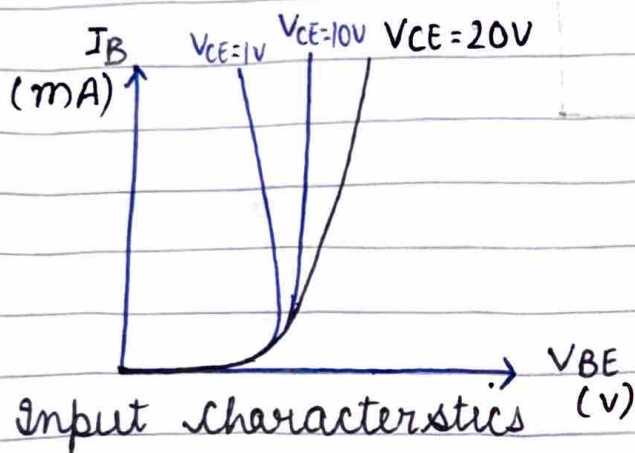
$$I_C = \alpha I_E + I_{CBO}$$

For common emitter

$$I_C = \beta I_B + (\beta + 1) I_{CBO}$$

large factor

Due to large $(\beta + 1)$ factor, contribution of leakage current I_{CBO} increases in common emitter configuration

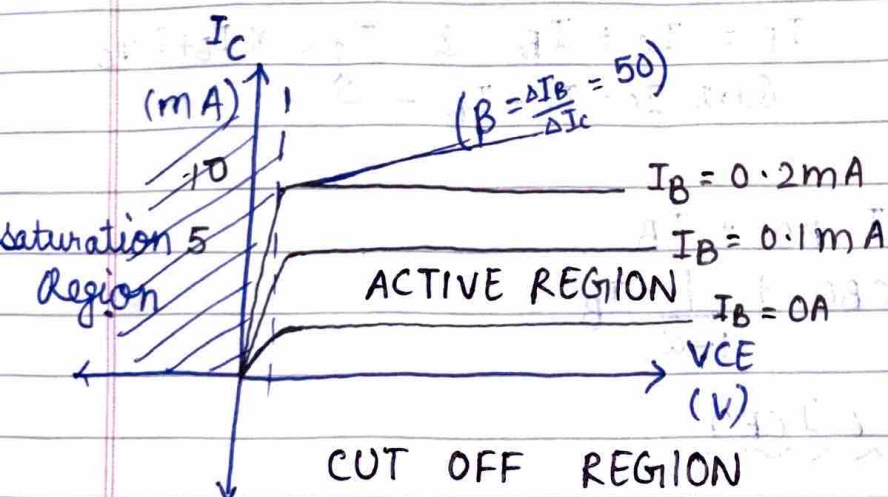


$$V_{CE} = V_{CB} + V_{BE}$$

$$V_{CE} \uparrow \Rightarrow V_{CB} \uparrow$$

$$\Rightarrow \text{width} \uparrow \Rightarrow W \uparrow$$

$$\Rightarrow W_{eff} \downarrow \Rightarrow I_B \downarrow$$



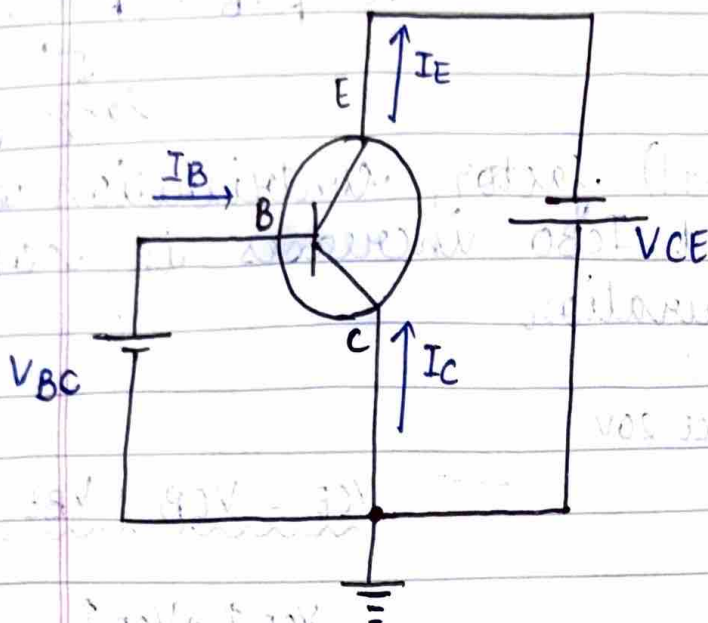
$$I_C = \beta I_B + (\beta + 1) I_{CBO}$$

$$I_B = 0$$

$$I_C = (\beta + 1) I_{CBO}$$

output characteristics

→ common collector configuration



$$I_C = \alpha I_E$$

$$\text{and } \alpha \approx 1$$

$$\Rightarrow I_C = I_E$$

$$\gamma = \frac{\Delta I_E}{\Delta I_B}$$

$$I_E = I_C + I_B \quad \& \quad I_C = \alpha I_E + I_{CBO}$$

$$\text{But } I_C = I_E - I_B \quad \rightarrow$$

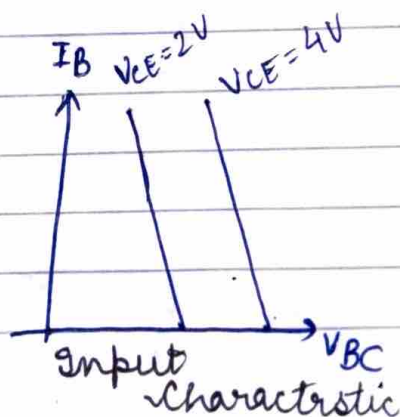
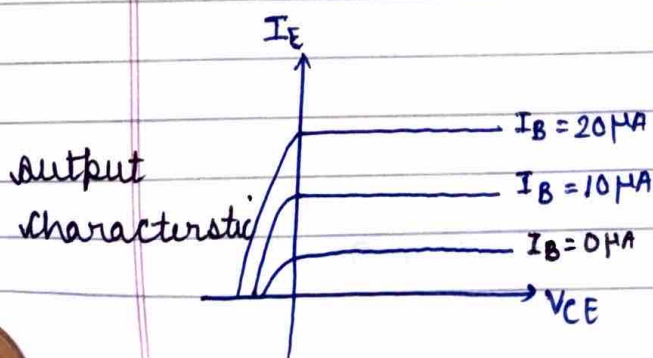
$$\Rightarrow I_E = \alpha I_E + I_{CBO} + I_B$$

$$\Rightarrow I_E = \frac{1}{1-\alpha} I_{CBO} + \frac{1}{1-\alpha} I_B$$

$$\Rightarrow I_E = \gamma I_B + \gamma I_{CBO}$$

$$I_{CBO} \ll I_B$$

$$\Rightarrow I_E = \gamma I_B$$



→ Relations b/t α, β, γ

α_{DC}

$$\alpha_{DC} = \frac{I_C}{I_E}$$

α_{AC}

$$\alpha_{AC} = \frac{\Delta I_C}{\Delta I_E} \bigg|_{V_{CB} \rightarrow \text{constant}}$$

//ly

$$\beta_{DC} = \frac{I_C}{I_B}$$

&

$$\beta_{AC} = \frac{\Delta I_C}{\Delta I_E} \bigg|_{V_{CE} \rightarrow \text{const}}$$

$$\gamma_{DC} = \frac{I_E}{I_B}$$

$$\gamma_{AC} = \frac{\Delta I_E}{\Delta I_B} \bigg|_{V_{CE} \rightarrow \text{constant}}$$

Since

$$I_E = I_C + I_B$$

$$\beta_{DC} = \frac{I_C}{I_B}$$

$$\gamma_{DC} = \beta_{DC} + 1$$

$$\Rightarrow \gamma = \beta + 1$$

$$\Rightarrow \gamma = \frac{\alpha + 1}{1 - \alpha}$$

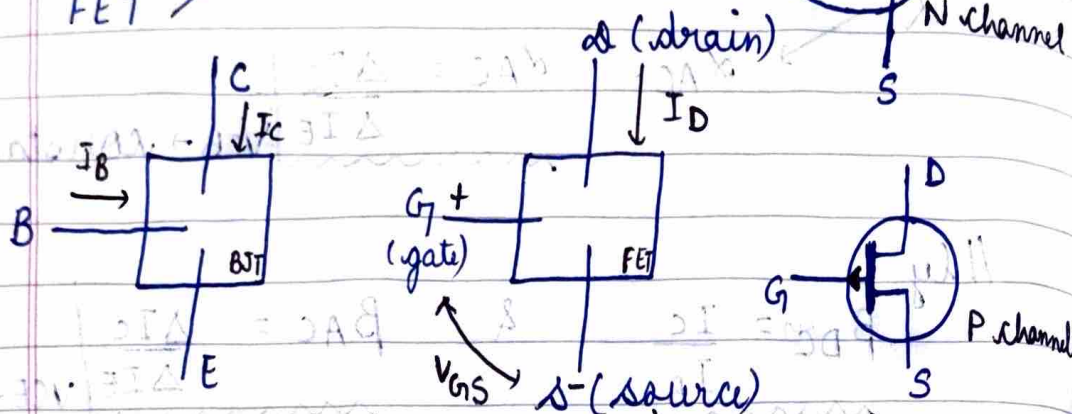
$$\Rightarrow \gamma = \frac{\alpha + 1 - \alpha}{1 - \alpha}$$

$$\Rightarrow \gamma = \frac{1}{1 - \alpha}$$

$$\gamma = \frac{1}{1 - \alpha} = \beta + 1$$

→ Field effect Transistors

BJT > 3 terminal devices
FET >



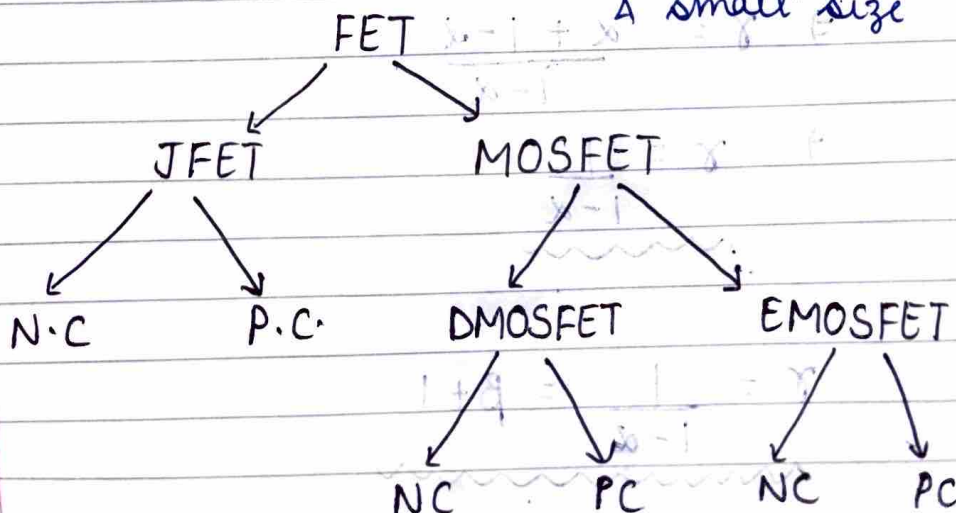
→ Both have nearly same applications

BJT

- Current controlled device ($I_C = \beta I_B$)
- Bipolar device
- conducts using both e^- and holes
- Low input impedance
- Large size
- Less temp. stability

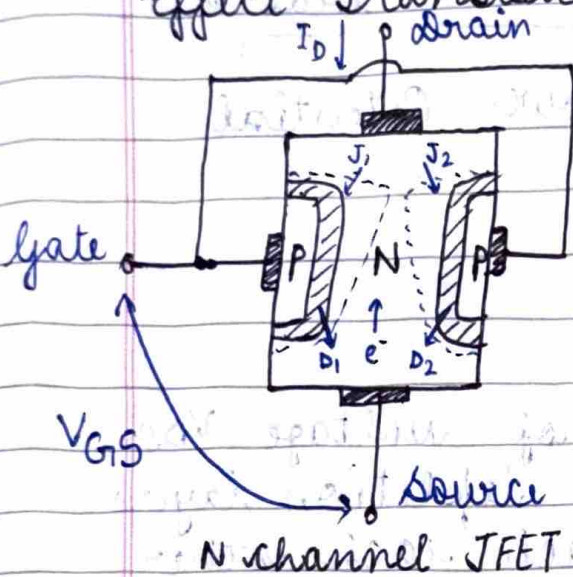
FET

- Voltage controlled device ($I_D = f(V_{GS})$)
- Unipolar device
- Used for amplification and switching
- High input impedance
- Uses only majority charge carriers
- More temp. stability
- A small size



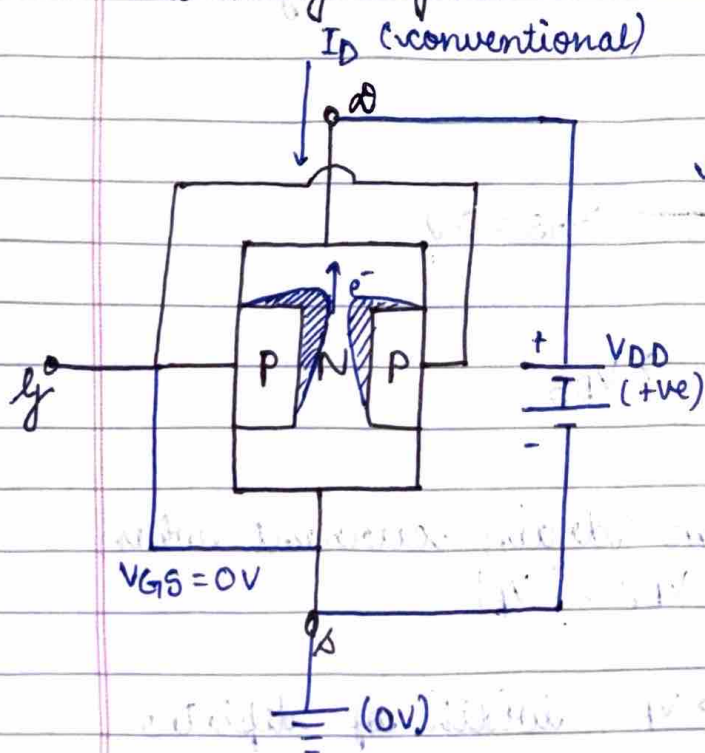
→ In FET, an electric field is developed by the charges present and this electric field controls the conduction path of output circuits.

→ Construction and Working of Junction Field Effect Transistor (JFET)



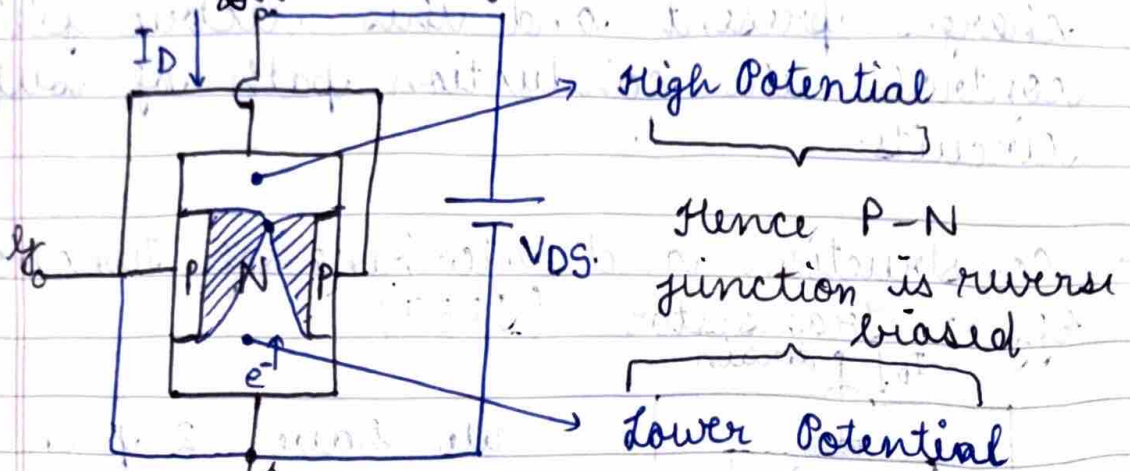
We have 2 p-n junctions J_1 and J_2 due to which 2 depletion layers are formed that are D_1 and D_2 .

→ Working of JFET ($V_{GS} = 0V$ and $V_{DS} > 0V$)



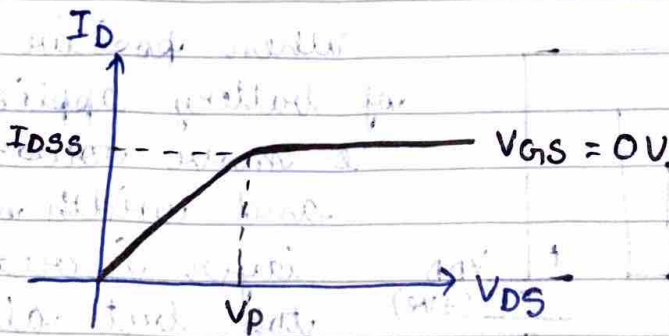
When positive terminal of battery applied at D , e^- move towards drain and width of depletion layer increases at top but almost same at bottom due to resistance of e^- flow by N channel.

→ Pinch off voltage (V_p)



$V_{DS} \uparrow$ $I_D \uparrow$ width $\uparrow$

The minimum value of voltage V_{DS} at which the two depletion layers of p-n junction ^{appears to} touch each other is called pinch off voltage and the current I_D becomes zero.



I_{DSS} is maximum drain current when $V_{GS} = 0V$ and $V_{DS} > |V_p|$

- when $V_{DS} \uparrow$ & $V_{DS} > V_p$, width of depletion region $\uparrow$ not at top but along the length of n channel and drain current remains same to I_{DSS} only when $V_{DS} < V_{DS \text{ max}}$
 $\hookrightarrow$ Breakdown

→ JFET as current source

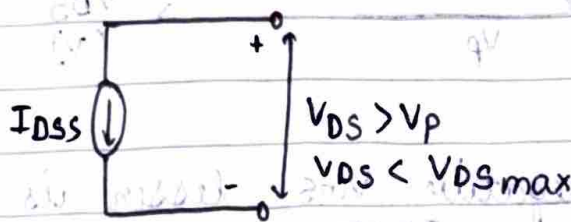
$$V_{DS} > V_P$$

$$I_D = I_{DSS}$$

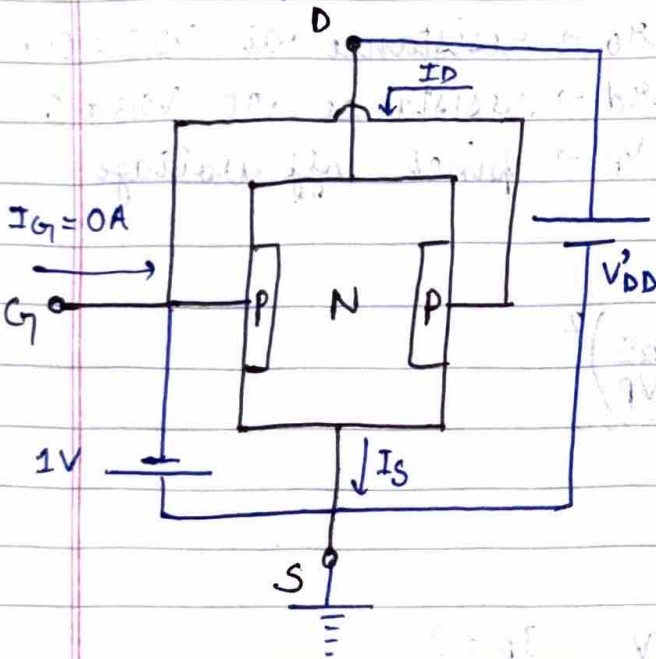
$$V_{DS} < V_{DS_{max}}$$

When these cond satisfies, the JFET works as current source

voltage at which breakdown of N-channel occurs.



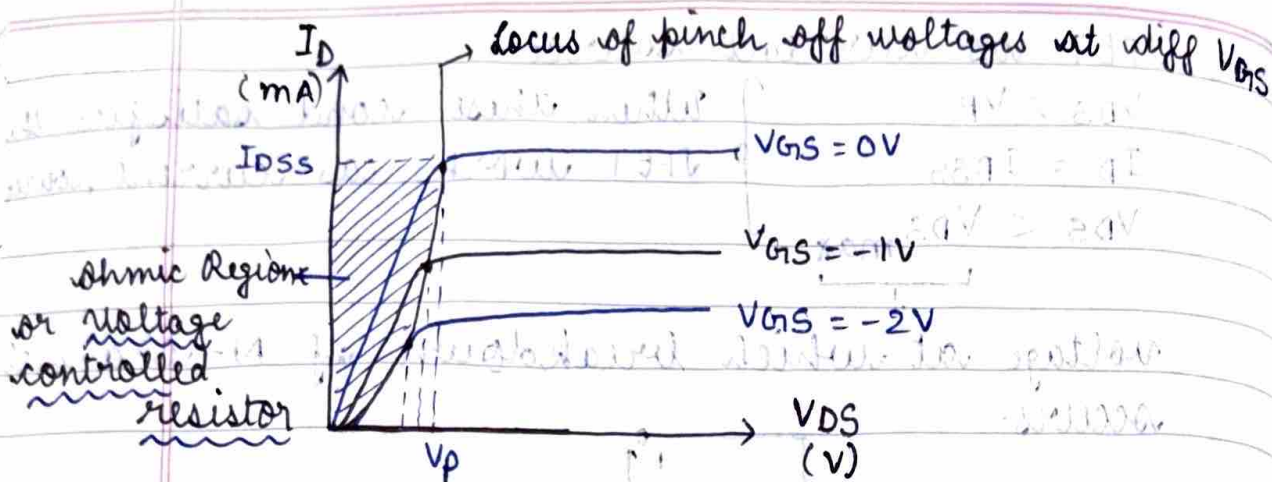
→ working of JFET ($V_{GS} < 0V$ & $V_{DS} = V_{DD} < V_{DD}$)
even when we change V_{DS} value of I_D remains constant ← to obtain saturation at lower value of V_{DS}



$V_{DS} \downarrow$ so to get same depletion width, we need to lower V_{GS} to a negative value

P type → lower potential
n type → higher potential
so both p-n junction are reverse biased

Negatively biasing gate gives pinch off at comparatively low voltage



More negative V_{GS} , lesser is V_p for N channel JFET

→ Resistance (JFET) = $f(V_{GS})$

$$\rightarrow R_d = \frac{R_0}{\left(1 - \frac{V_{GS}}{V_p}\right)^2}$$

R_0 → resistance at $V_{GS} = 0V$

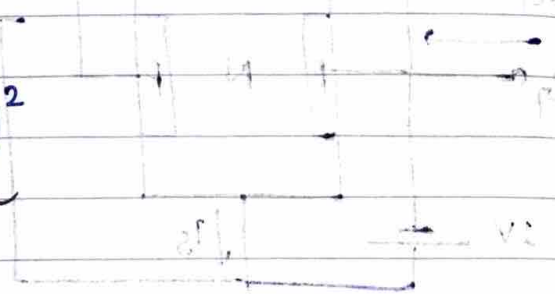
R_d → resistance at $V_{GS} \neq 0$

V_p → pinch off voltage

→ Shockley's equation

$$I_D = I_{DSS} \left(1 - \frac{V_{GS}}{V_p}\right)^2$$

constant



Q. N channel JFET.

$$I_{DSS} = 10 \text{ mA} \quad V_p = -4 \text{ V} \quad I_D = ?$$

a) $V_{GS} = 0 \text{ V}$

b) $V_{GS} = -1 \text{ V}$

c) $V_{GS} = -4 \text{ V}$

a) $I_D = 10 \left(1 - \frac{0}{-4}\right)^2 = 10 \text{ mA}$

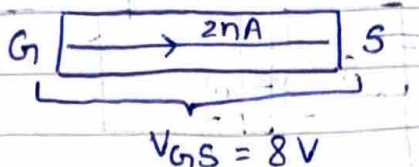
b) $I_D = 10 \left(1 - \frac{-1}{-4}\right)^2 = \frac{10^5 \times 9}{168} = 5.625 \text{ mA}$

c) $I_D = 0 \text{ mA}$

- Q. A JFET produces a gate current of 2 nA when gate is reverse biased with 8 V . Determine resistance between gate & source.

$$I_G = 2\text{ nA}$$

$$V_{GS} = 8\text{ V}$$



$$V = IR$$

$$R_{GS} = \frac{V_{GS}}{I_G} = \frac{8\text{ V}}{2 \times 10^{-9}\text{ A}} = 4 \times 10^9 \Omega = 4000\text{ M}\Omega$$

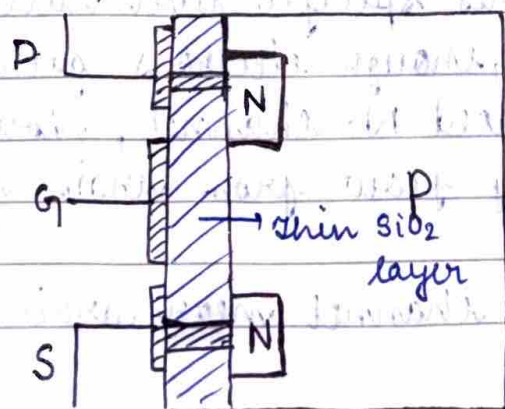
- Q. Reverse ^{gate} voltage of JFET changes from 4.4 to 4.2 V & drain current from 2.2 mA to 2.6 mA . Find out transconductance $\rightarrow g_m = \frac{\text{o/p current}}{\text{i/p voltage}} \bigg|_{\text{o/p V}}$
- $V_{GS} : 4.4 \rightarrow 4.2$
- $I_D : 2.2 \rightarrow 2.6$

$$\text{transconductance} = g_m = \frac{\Delta I_D}{\Delta V_{GS}} \bigg|_{\text{constant } V_{DS}}$$

$$g_m = \frac{2.2 - 2.6}{4.4 - 4.2} = \frac{0.4}{0.2} = 2\text{ m mho}$$

→ construction and working of enhancement type metal oxide semiconductor field effect transistor (EMOSFET)

MOSFET has ability to control current flow, so it's an active device. eg → MOSFET, JFET, BJT

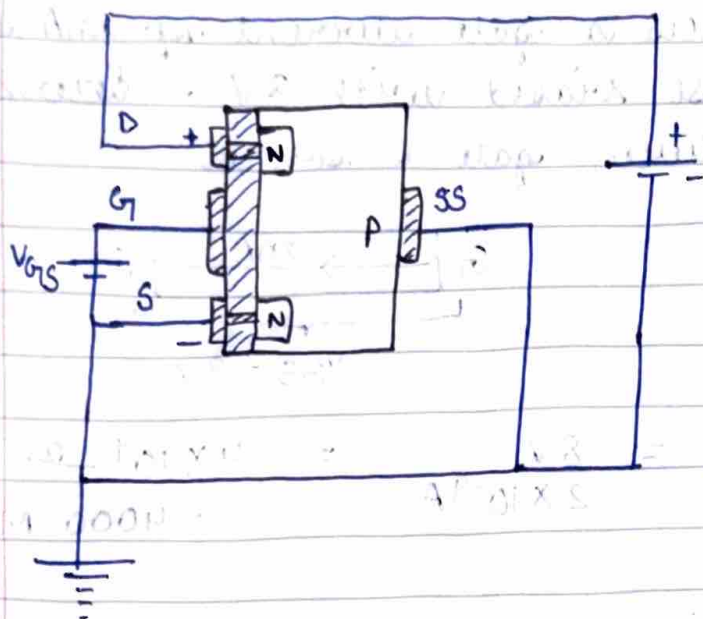


n-channel EMOS

Initially there is no channel between D & S

in EMOS.

major → holes minor → e^-



E. mosfet consists of a p-type substrate with two highly doped N-type regions i.e. S & D. The gate is a thin layer of SiO_2 which acts as insulator & creates capacitor type structure.

→ Working

• OFF state ($V_{DS} = 0V$)

When $V_{GS} = 0$, there is no channel; even if we apply V_{DS} , no current flows because the P substrate acts like open switch.

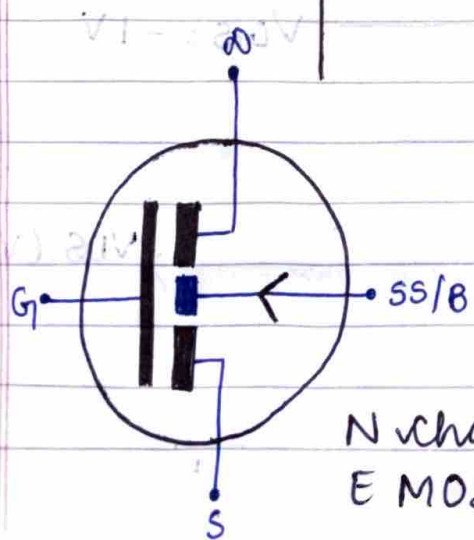
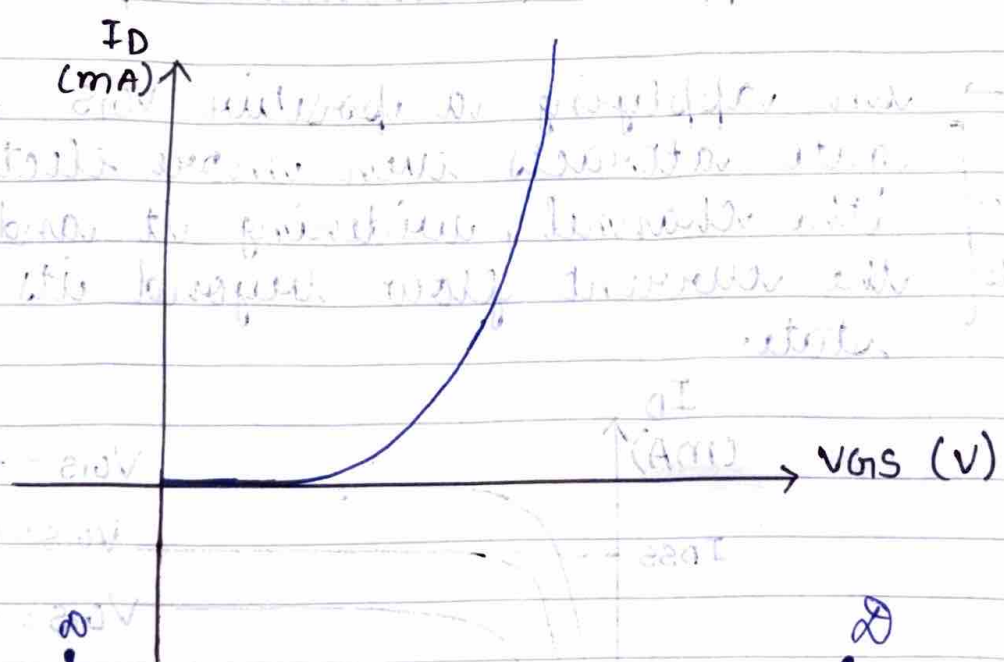
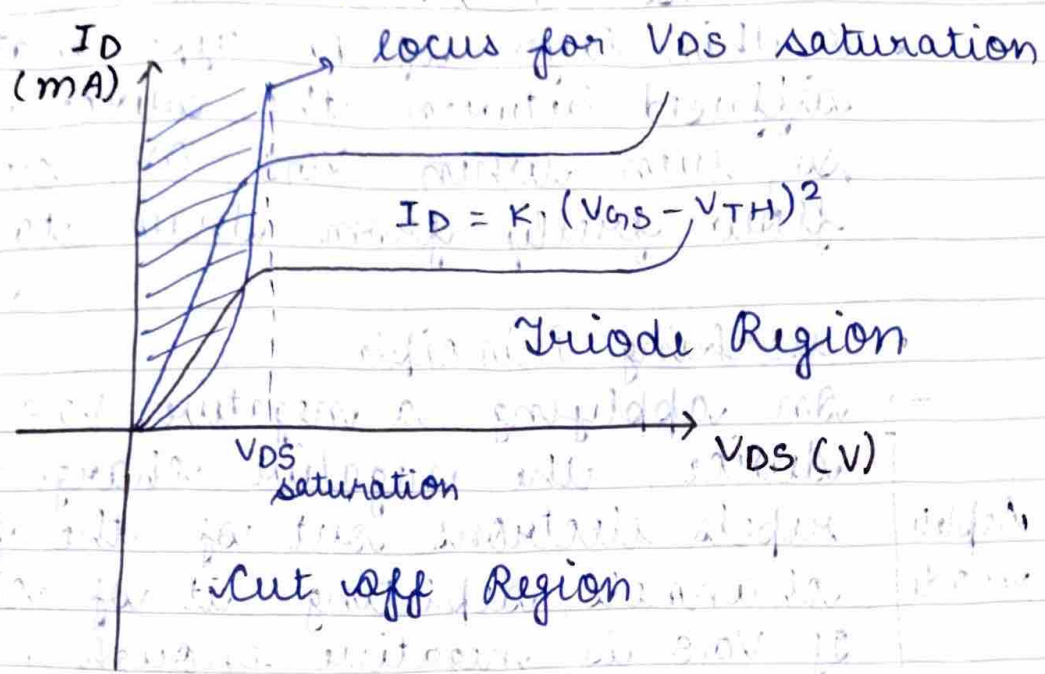
• CREATING CHANNELS ($V_{DS} > 0V$)

As we increase V_{GS} and make it positive, the charge on gate pushes away holes in the P substrate and attracts electrons towards the surface.

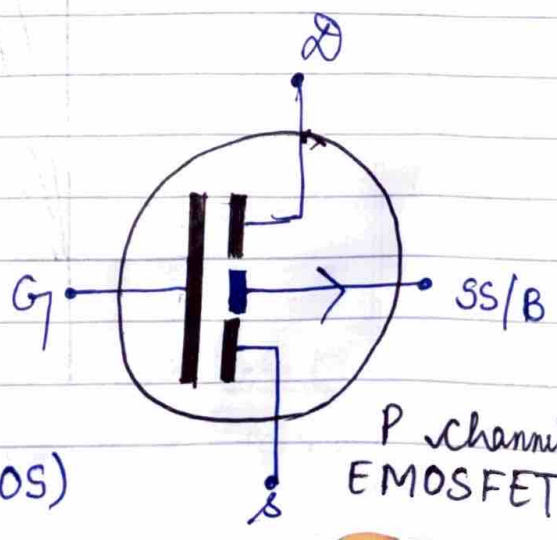
Once V_{GS} reaches specific level called as threshold voltage (V_{TH}), enough electrons gather to form a bridge / induced N-channel. Now the current can finally flow from drain to source.

When $V_{DS} = V_{GS} - V_{TH}$, channel near drain will

become extremely narrow and width will increase very much and pinch off arrives



N channel
E MOSFET (NMOS)



P channel
E MOSFET

→ Depletion Type MOSFET

It has a built channel between drain and source unlike EMOS

- A thin layer of N type material is diffused between the source and drain so even when $V_{GS} = 0V$, current can flow freely from drain to source.

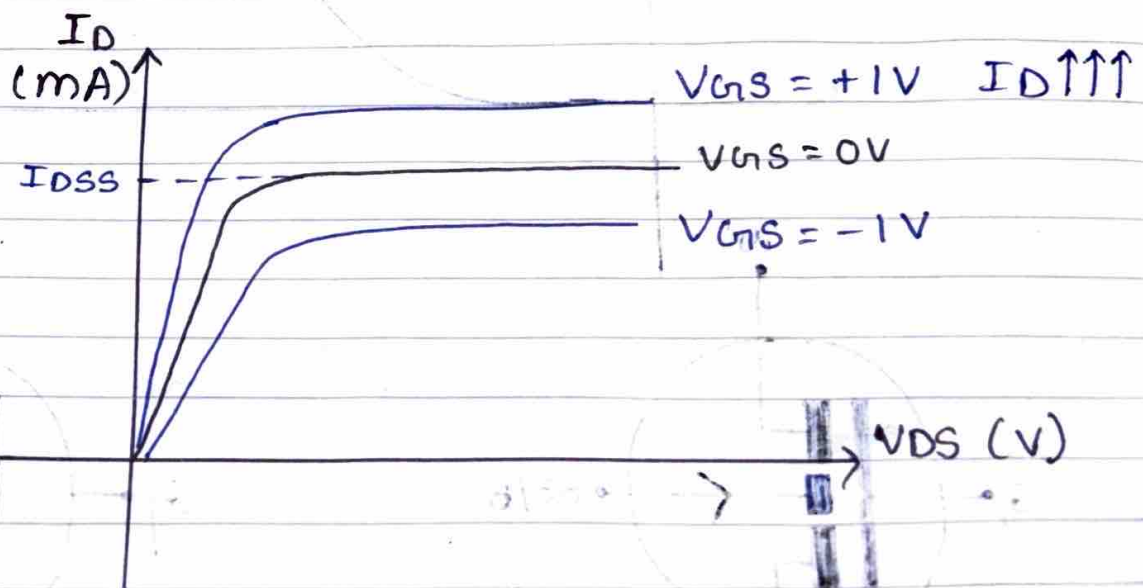
• Working Principle

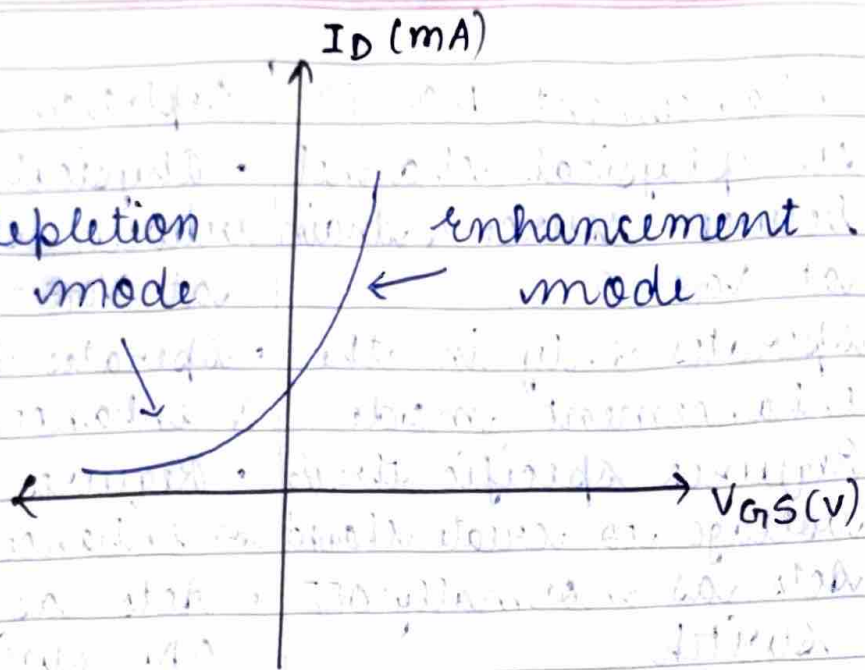
→ In applying a negative V_{GS} to N channel device, the negative charge on the gate repels electrons out of the pre-built channels, depleting it of charge carriers. If V_{GS} is negative enough, the channel disappears & current stops.

depleti
mode

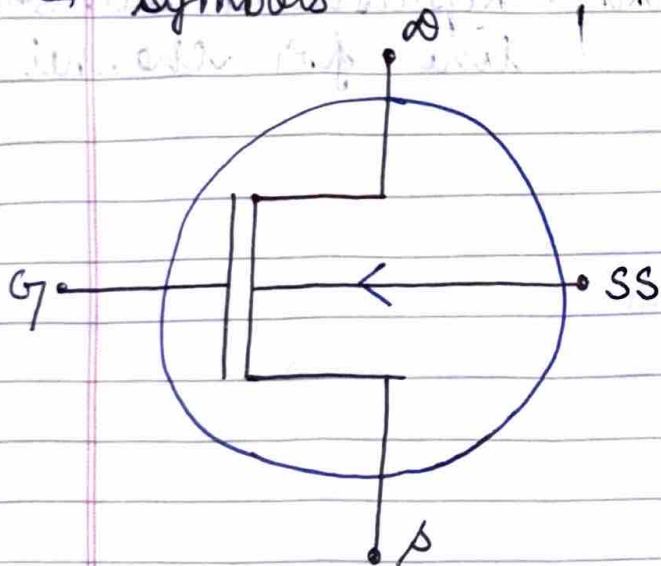
→ In applying a positive V_{GS} , the positive gate attracts even more electrons into the channel, widening it and increasing the current flow beyond its natural state.

enhance
mode

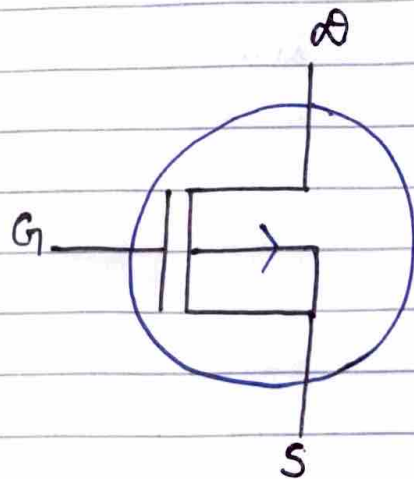




→ Symbols



N channel
MOSFET



P channel MOSFET
(PMOS)

→

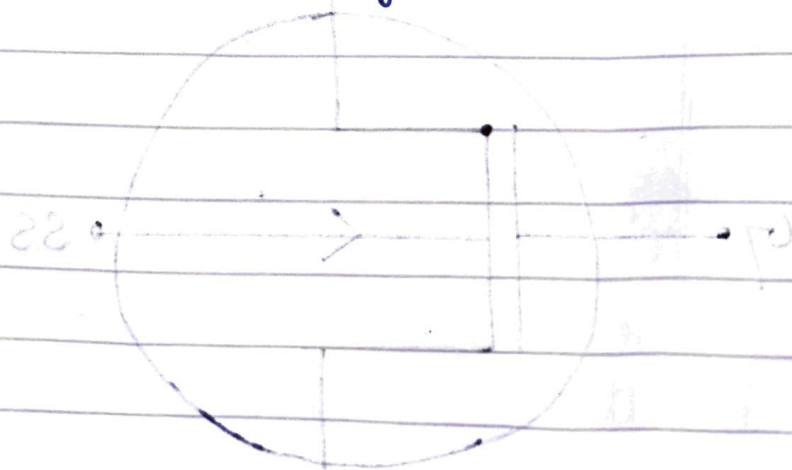
(A.M.) a.I

enhancement MOSFET

- No physical channel between source & drain at $V_{GS} = 0$
- Operates only in the enhancement mode
- Requires specific threshold voltage to create channel
- Acts as normally OFF switch
- $I_D = 0$ at $V_{GS} = 0$
- Represented with broken line for channel

Depletion MOSFET

- Physical channel exists between source & drain at $V_{GS} = 0V$
- Operates in both depletion & enhancement modes
- Requires voltage to deplete or enhance existing carriers
- Acts as normally ON switch
- I_D flows even when $V_{GS} = 0$
- Represented with solid line for channel



unit → 2 Remaining

→ workings of transistors

1. CE BJT

The emitter is shared between base and collector. The base-emitter junction is forward biased while collector-emitter junction is reverse biased. When small input voltage is applied to base, it overcomes barrier potential of base-emitter junction, allowing electrons to flow from heavily doped emitter into the base region but base is very lightly doped so only a tiny fraction of these electrons recombine with holes in base to form base current I_B and remaining i are attracted by reverse biased collector voltage which creates a much larger collector current.

2. CB BJT

Base is earthed & serves as common terminal between emitter and collector. The emitter-base junction is forward biased and collector-base junction is reverse biased. Because the emitter is heavily doped, forward bias cause large no of majority charge carriers to inject into base and enter a thin lightly doped region and a very small percentage recombine to form base current and rest majority is immediately collected by collector due to reverse biasing and leads to produce I_C which is slightly less than I_E .

→ CC BJT

collector-base junction is reverse biased and base-emitter junction is forward biased. The input signal is applied at base while output signal is taken from emitter. Working is based on fact that emitter voltage follows input base voltage minus small forward bias voltage drop. When $V_B \uparrow$, forward bias on base-emitter junction $\uparrow$ causing more current to flow from collector through emitter, this creates larger voltage drop and increases output voltage.

→ C-MOSFET

complementary Metal oxide semiconductor field effect transistor.

It's used in modern IC's and combines both PMOS and NMOS enhancement mode MOSFETs in pairs to perform logic functions.

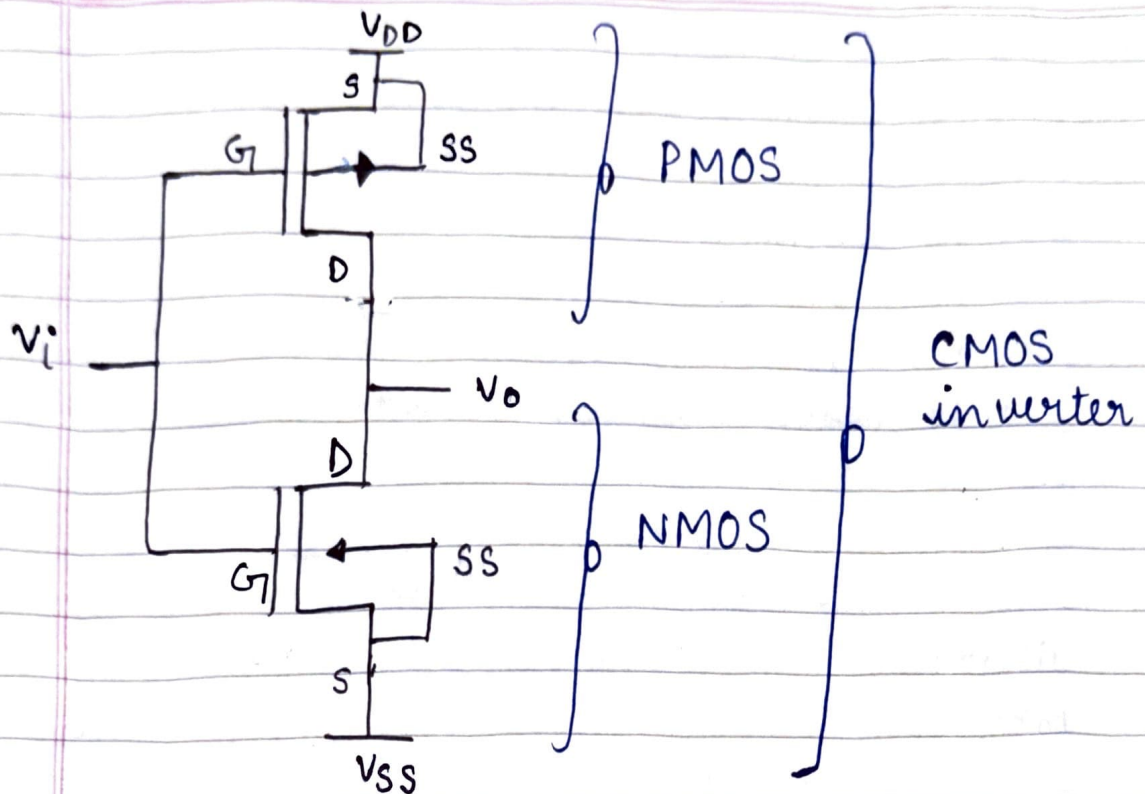
Structure

PMOS & NMOS are typically connected in series between drain supply voltage and source ground voltage (V_{SS}) (V_{DD}).

→ PMOS only turns on when input is low and is always connected to V_{DD} .

→ NMOS only turns on when input is high & is connected to V_{SS} .

eg → CMOS inverter



It toggles the input as output

- When input is low, PMOS is switched on and NMOS is switched off which pulls up output terminal V_o resulting in high output
- When input is high, PMOS switches OFF and NMOS is switched on which pulls down output terminal to ground resulting in low output.